

Name : Arthi. V  
College : VSB Engineering college technical campus  
E-mail : [varthi190@gmail.com](mailto:varthi190@gmail.com)  
Topic Natural language generator

## **NATURAL LANGUAGE GENERATOR**

### **TAB 1:** Introduction

Natural Language Generation (NLG) is a branch of AI.  
It converts structured data into human-readable text.  
Used in chatbots, report generation, and virtual assistants.  
Helps automate content creation.

### **TAB 2:** Problem Statement

Problem

Creating text manually is time-consuming.  
Large amounts of data need quick summaries.  
Businesses require automated report generation.

Solution

Develop a Natural Language Generator using Python.

### **TAB 3:** Objectives

Project Objectives

Generate meaningful sentences automatically.  
Reduce manual writing effort.  
Improve communication between humans and machines.  
Demonstrate AI-based text generation.

### **TAB 4:** System Architecture

Architecture

Input Data → Processing → Text Generation → Output Text

Modules:

User Input  
Text Processing  
Sentence Generation  
Output Display

### **TAB 5:** Technologies Used

Tools & Technologies

Python  
Natural Language Processing (NLP)  
NLTK Library  
VS Code / PyCharm  
Windows Operating System

### **TAB 6:** Algorithm

Steps

Receive user input.  
Process text using NLP.  
Analyze keywords.  
Generate appropriate sentences.  
Display generated text.

#### **TAB 7: Python Code**

Python

```
responses = {  
    "hello": "Hello! How are you?",  
    "python": "Python is a popular programming language.",  
    "ai": "Artificial Intelligence helps machines think."  
}  
  
user = input("Enter a word: ").lower()  
  
if user in responses:  
    print(responses[user])  
else:  
    print("Sorry, I don't know about that.")
```

#### **TAB 8: Advantages**

Advantages  
Easy to use  
Fast text generation  
Reduces manual effort  
Improves productivity  
Supports automation

#### **TAB 9: Applications**

Applications  
Chatbots  
Virtual Assistants  
Content Creation  
Report Generation  
Customer Support  
Future Scope  
Integrate Machine Learning  
Support multiple languages  
Voice-based interaction  
Advanced AI text generation

#### **TAB 10: Conclusion**

Conclusion  
Natural Language Generation is an important AI technology.  
It automates text creation efficiently.  
The project demonstrates basic NLG concepts using Python.

**THANK YOU ☺**